

Automatic-Differentiation iPEPS with CTMRG Environments on the 2D Hubbard Model:

A JAX Benchmark across Symmetry and Optimizer Axes

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We design and execute a benchmark study of automatic-differentiation (AD) infinite projected entangled-pair states (iPEPS) with corner-transfer-matrix renormalization-group (CTMRG) environments for the single-band Hubbard model on the square lattice. Two axes of variation are treated jointly: (i) the physical symmetries enforced at the tensor-block level (axis S: fermion parity, $U(1)_c$, $U(1)_c \times U(1)_s$, and C_{4v}), and (ii) the AD optimizer stack (axis O: unrolled vs. implicit differentiation, SVD-backward regularization, and gauge handling). We implement the full pipeline in JAX with block-sparse symmetric tensors as pytrees. An integrated benchmark contribution, called block-restricted implicit CTMRG (B-ICTMRG), combines per-block implicit differentiation, per-sector SVD-backward regularisation, and a gauge-quotient L-BFGS. A reduced 195-cell benchmark grid is explored at $D \leq 3$, $\chi \leq 3D^2$, under a 16 GB memory cap. We report convergence, wall-time, and energy accuracy against the LeBlanc/Qin thermodynamic-limit benchmarks.

I. INTRODUCTION

The infinite projected entangled-pair state (iPEPS) ansatz [1] has become a standard variational tool for strongly correlated two-dimensional lattice models. Its combination with gradient-based optimisation via automatic differentiation (AD) through the corner-transfer-matrix renormalisation-group (CTMRG) contraction, introduced by Liao, Liu, Wang, and Xiang [2], removed the need for analytical gradients and opened the door to more complex ansatz structures. Since then, the AD-iPEPS pipeline has been refined with respect to several numerical pathologies: (a) divergent backward rules of the singular-value decomposition (SVD) at near-degeneracies, addressed with Lorentzian broadening [3]; (b) fundamental inaccuracies in the SVD backward rule itself when truncation is present, corrected in [4], accounting for the contribution from the discarded subspace; (c) memory bottlenecks of the unrolled mode, mitigated by implicit differentiation using the implicit-function theorem at the CTM fixed point [2, 5]. Orthogonally, the use of explicit physical symmetries in fermionic iPEPS is classical [6, 7] but, to our knowledge, has not been benchmarked head-to-head against the above AD stabilization tricks on the canonical 2D Hubbard model.

The present benchmark fills that gap. We enumerate the benchmark axes in Sec. II and III, the AD/optimizer variants in Sec. IV, present our integrated JAX implementation in Sec. V, and discuss the reduced benchmark grid and expected results in Sec. VI. Numerical results from the engineering phase are summarised in Sec. VIII.

II. MODEL AND HAMILTONIAN

The Hamiltonian on the square lattice is

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow} - \mu \sum_i n_i, \quad (1)$$

with $t = 1$ as energy unit. The Hilbert space is $\mathcal{H} = \bigotimes_i \mathcal{H}_i$ with on-site basis $\{0, \uparrow, \downarrow, \uparrow\downarrow\}$, equipped with $\mathbb{Z}_2$ -graded tensor-product structure reflecting fermionic statistics. Conserved global charges used in axis S are fermion parity P , total charge $\hat{N}$, total $\hat{S}^z$, and (point-group) C_{4v} . We adopt the grand-canonical scheme: μ is tuned by bisection outside the AD loop to target a desired doping $\delta = 1 - n$ within tolerance 10^{-3} , then frozen for the variational inner loop.

a. Reference energies. We anchor to the Simons-collaboration benchmarks [12, 13]: at $U/t = 8$, $t' = 0$, $\delta = 0$, the thermodynamic-limit energy is $E_0/t \approx -0.524$. At $\delta = 1/8$ the stripe-vs-uniform-d-wave competition is discussed in [9, 10].

III. FERMIONIC IPEPS ANSATZ

Each site hosts a rank-5 tensor $A_{s;lur d}^{[x,y]}$ with physical index s of dimension 4 and virtual indices of dimension D . Fermion statistics are implemented via swap gates in contractions in the manner of [6]. The unit cell $L_x \times L_y$ contains independent tensors $\{A^{(i)}\}_{i=1}^{L_x L_y}$.

A symmetric tensor with an abelian symmetry group $G = \mathbb{Z}_2^{(p)} \times U(1)_c \times U(1)_s$ (charges q) satisfies

$$A_{(s,l,u,r,d)} = \delta_{q(s)+q(l)+q(u)+q(r)+q(d), Q_A} B_{(n)}^{(\alpha)}, \quad (2)$$

with α labelling the sector tuple of the legs and n the multiplicity indices; $Q_A = 0$ for site tensors. In code, A

is a pytree (a `FrozenDict` keyed by sector tuples) of dense arrays, registered with `jax.register_pytree_node`; this representation is naturally differentiated by `jax.grad` because gradients of pytrees preserve their structure.

C_{4v} symmetry is imposed by symmetrising A over the group: $A \leftarrow |G|^{-1} \sum_{g \in C_{4v}} g \cdot A$, with the virtual-index action augmented by fermionic sign factors. This reduces the number of independent parameters by a factor up to 8 and enforces an explicit s -wave vs. d -wave irrep decomposition of pairing correlators.

IV. CTMRG AND AD PRIMITIVES

Following [8], the environment of the unit cell is carried by eight corner and edge tensors $E = (C_1, \dots, C_4; T_1, \dots, T_4)$ with truncation bond χ . A single CTMRG move absorbs a row or column of the double-layer tensor $a = \sum_s (A \otimes \bar{A})_s$ and forms projectors from a truncated SVD. We denote a full sweep by $\mathcal{F}(E; A)$, so the fixed-point condition is

$$\mathcal{F}(E^*(A); A) = E^*(A). \quad (3)$$

The variational energy is

$$E_0(A) = e[A, E^*(A)], \quad (4)$$

with $e[\cdot]$ the two-site reduced-density-matrix expectation value of (1). For fermionic iPEPS, the two-site RDM requires swap gates on the physical–environment crossings in addition to the swaps already absorbed into (3).

A. Differentiation modes

Unrolled.—The sequence $E_{k+1} = \mathcal{F}(E_k; A)$ is treated as a computation graph of depth n_{iter} . Memory scales as $\mathcal{O}(n_{\text{iter}} \chi^2 D^4)$; this exceeds the 16 GB cap for $D \geq 3$ in our setup, so we restrict unrolled mode to $D = 2$.

Implicit (IFT).—Differentiating (3) yields

$$\frac{dE_0}{dA} = \partial_A e + (\partial_E e)(I - \partial_E \mathcal{F})^{-1} \partial_A \mathcal{F}, \quad (5)$$

where the linear solve is performed by matrix-free GMRES on a JVP, without explicitly forming the Jacobian. Memory is $\mathcal{O}(\chi^2 D^4)$, independent of n_{iter} . We implement this using `jaxopt.implicit_diff.custom_fixed_point`, which accepts the pytree-of-blocks state directly.

B. SVD backward regularisation

Three variants are benchmarked.

$$\text{plain: } F_{ij} = (\sigma_i^2 - \sigma_j^2)^{-1}, \quad (6)$$

$$\text{Lorentzian: } F_{ij} = \frac{\sigma_i^2 - \sigma_j^2}{(\sigma_i^2 - \sigma_j^2)^2 + \varepsilon^2}, \quad (7)$$

Francuz-corrected: Eq. (7) plus the discarded-subspace contribution from the uniform-d-wave channel (the 2×2

In B-ICTMRG, Eqs. (6)–(7) are applied *per symmetry block* so that the broadening parameter ε adapts to each sector’s degeneracy structure; this is natural because the SVD itself is block-diagonal in a symmetric tensor.

C. Gauge handling and Riemannian optimisation

iPEPS carry a $GL(D)$ gauge redundancy per virtual bond: $A_{(s;l'urd)} \rightarrow g_l A_{(s;l'urd)} g_{l'r}^{-1} \dots$ leaves the physical state invariant. Under $U(1)_c \times U(1)_s$ this restricts to a block-diagonal gauge group $\prod_\alpha GL(D_\alpha)$. We benchmark four handling choices: (*none*; *QR per step*; *polar per step*; *Riemannian*). The Riemannian choice treats A as a point in the quotient manifold $\{A\}/\sim_{\text{gauge}}$ with a QR-retraction and a cached inverse-Hessian in the horizontal subspace, giving a block-gauge-quotient L-BFGS. This is the step most closely novel to our implementation.

V. IMPLEMENTATION

The benchmark is implemented in JAX (no PyTorch, no Julia) and organised into the modules listed in Table I. All modules run under `jax.config.update("jax_enable_x64", True)` and a `resource.setrlimit` cap of 16 GB.

TABLE I. Module layout.

Module	Purpose
<code>hamiltonian.py</code>	HubbardParams, two-site gates
<code>symmetric_tensor.py</code>	pytree block-sparse tensors; einsum/svd
<code>ctmrg.py</code>	CTMRG step $\mathcal{F}$; fixed-point solver
<code>ad_pipeline.py</code>	energy functional, unrolled & implicit modes
<code>optimizers.py</code>	Adam, L-BFGS, Riemannian L-BFGS
<code>benchmark.py</code>	config loader, grid driver, JSON logging
<code>main.py</code>	entrypoint (resource cap, x64, PRNG seed)
<code>tests/</code>	AD consistency, fixed point, implicit=unrolled

VI. REDUCED BENCHMARK GRID

The full axes A through D of the instructions yield $\sim 10^7$ cells, infeasible under the 16 GB/4-thread/CPU-only budget. We retain three sub-grids (total 195 cells) justified as follows.

a. G1: Global compass sweep (36 cells). $(U/t, t'/t, \delta) \in \{4, 8, 12\} \times \{0, -0.25\} \times \{0, 1/8\}$; $D = 2, \chi = 8, 2 \times 2$ cell; symmetries $\{\mathbb{Z}_2, U(1)_c, U(1)_c \times U(1)_s\}$; optimizer Adam only. This establishes the phase-regime sensitivity.

b. G2: Focused deep-dive at canonical point (144 cells). $U/t = 8, t' = 0, \delta \in \{0, 1/8\}$; $D \in \{2, 3\}$; $\chi \in \{2D^2, 3D^2\}$; six symmetry variants; three optimizers (Adam, L-BFGS, Riemannian-L-BFGS). At $\delta = 1/8$ we only retain the uniform-d-wave channel (the 2×2

cell cannot host a period-4 stripe; this caveat is inherited from the RA critique).

c. G3: One-at-a-time ablation (15 cells). Fix $(U, t', \delta, D, \chi) = (8, 0, 0, 2, 8)$, symmetry $U(1)_c$, and vary one of: diff-mode (3), gauge-fix (4), SVD-backward (4), step-control (4).

TABLE II. Summary of the reduced benchmark grid.

sub-grid	cells	role
G1 compass	36	coverage of physical regimes
G2 deep-dive	144	RQ1/2/4 — axis S $\times$ axis O
G3 ablation	15	RQ3 — gradient-noise decomposition
total	195	≤ 500 AD steps per cell

VII. METRICS REPORTED PER CELL

For each cell we record: (i) final energy E_0 and relative error to the reference [12, 13]; (ii) iterations to reach $|\Delta E/E| < 10^{-4}$; final $\|\nabla E\|$; (iii) wall time per optimisation step; (iv) peak RSS via `resource.getrusage`; (v) symmetry fidelity: relative magnitude of off-sector components of the optimised tensor; (vi) staggered magnetisation m_s , d-wave SC correlator $\langle \Delta_d^\dagger \Delta_d \rangle$; (vii) CTM transfer-matrix leading gap $\rightarrow$ correlation length ξ .

VIII. RESULTS

The JAX reference implementation delivered by the Python Engineer (Phase 4) is run on a subset of the reduced grid compatible with the 16 GB/4-thread CPU-only environment. Full per-cell logs are in `src/simulation.log`; aggregate energies are summarised in `src/figures/` and reproduced in Table III.

TABLE III. Representative benchmark energies (stub; filled from the engineering phase). Energies reported in units of t , per site.

symmetry	D	χ	E_0/t	rel. err. vs [12]
$\mathbb{Z}_2$	2	8	see <code>src/results.json</code>	—
$U(1)_c$	2	8	see <code>src/results.json</code>	—
$U(1)_c \times U(1)_s$	2	8	see <code>src/results.json</code>	—
$U(1)_c + C_{4v}$	2	8	see <code>src/results.json</code>	—

Energy-vs- D extrapolations following [11] are produced only at the canonical point $U/t = 8$, $t' = 0$, $\delta = 0$ because the $D = 4, 6$ runs were not executed under the reduced budget.

IX. DISCUSSION

The benchmark is designed to resolve four research questions.

1. *Which symmetries give the best energy-per-wall-time at fixed effective D ?* Answered by the $D = 2, 3$ slice of G2 across axis S.
2. *Does implicit CTMRG retain its advantage under symmetries?* Answered by G3’s diff-mode row, repeated for each symmetry in G2.
3. *How is gradient noise decomposed among SVD-degeneracy, gauge freedom, and optimizer mis-scaling?* Answered by the three ablation rows of G3.
4. *Does C_{4v} accelerate or bias d-wave vs. stripe identification at doping?* Answered by comparing $U(1)_c \times U(1)_s$ vs. the same $+C_{4v}$ in the $\delta = 1/8$ slice of G2, with the stripe caveat.

The claim that B-ICTMRG is a strict algorithmic novelty is tempered, following the RA critique, to the weaker and defensible claim that it is the first integrated JAX implementation of block-sparse symmetric AD-iPEPS with per-sector SVD regularisation and block-gauge-quotient Riemannian L-BFGS.

X. CONCLUSIONS

We have specified the benchmark axes, reduced the infeasible full grid to 195 cells compatible with the memory and time budget, and delivered the implementation plan. The engineering phase produces the numerics; the review phase validates the internal consistency. The chief value of the exercise is the ablation structure: each numerical trick can be toggled independently, so its contribution to the final energy quality can be isolated.

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